Newman-Penrose Formalism Calculations

Stephani et al., Equation (15.3_k0)

Metric

$$ds^2 = -x_1^2 Y\left(x_3, x_4\right)^2 \mathrm{d}x_2 \otimes \mathrm{d}x_2 + Y\left(x_3, x_4\right)^2 \mathrm{d}x_1 \otimes \mathrm{d}x_1 + e^{(2\,\lambda(x_3,x_4))}\mathrm{d}x_3 \otimes \mathrm{d}x_3 + e^{(2\,\nu(x_3,x_4))}\mathrm{d}x_4 \otimes \mathrm{d}x_4$$

Coordinates

$$(x^\mu) = (x_2, x_1, x_3, x_4)$$

Metric Functions

$$\Sigma = x_1$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2}\sqrt{2}x_1 Y\left(x_3, x_4\right)\right) dx_2 + \left(-\frac{1}{2}\sqrt{2}Y\left(x_3, x_4\right)\right) dx_1$$

$$n = \left(\frac{1}{2}\sqrt{2}x_1 Y\left(x_3, x_4\right)\right) dx_2 + \left(\frac{1}{2}\sqrt{2}Y\left(x_3, x_4\right)\right) dx_1$$

$$m = \left(\frac{1}{2}\sqrt{2}e^{(\lambda(x_3,x_4))}\right) dx_3 + \left(\frac{1}{2}i\sqrt{2}e^{(\nu(x_3,x_4))}\right) dx_4$$

$$\overline{m} = \left(\frac{1}{2}\sqrt{2}e^{(\lambda(x_3,x_4))}\right) dx_3 + \left(-\frac{1}{2}i\sqrt{2}e^{(\nu(x_3,x_4))}\right) dx_4$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}}{2\,x_1 Y\left(x_3, x_4\right)}\right)\,dx_2 + \left(-\frac{\sqrt{2}}{2\,Y\left(x_3, x_4\right)}\right)\,dx_1$$

$$n = \left(-\frac{\sqrt{2}}{2\,x_1 Y\left(x_3, x_4\right)}\right)\,dx_2 + \left(\frac{\sqrt{2}}{2\,Y\left(x_3, x_4\right)}\right)\,dx_1$$

$$m = \left(\frac{1}{2}\,\sqrt{2}e^{\left(-\lambda\left(x_3, x_4\right)\right)}\right)\,dx_3 + \left(\frac{1}{2}i\,\sqrt{2}e^{\left(-\nu\left(x_3, x_4\right)\right)}\right)\,dx_4$$

$$\overline{m} = \left(\frac{1}{2}\,\sqrt{2}e^{\left(-\lambda\left(x_3, x_4\right)\right)}\right)\,dx_3 + \left(-\frac{1}{2}i\,\sqrt{2}e^{\left(-\nu\left(x_3, x_4\right)\right)}\right)\,dx_4$$

Directional Derivatives

$$D = -\frac{\sqrt{2}}{2\,x_1 Y\left(x_3, x_4\right)}\,\partial_{x_2} - \frac{\sqrt{2}}{2\,Y\left(x_3, x_4\right)}\,\partial_{x_1}$$

$$\Delta = -\frac{\sqrt{2}}{2\,x_1 Y\left(x_3, x_4\right)}\,\partial_{x_2} + \frac{\sqrt{2}}{2\,Y\left(x_3, x_4\right)}\,\partial_{x_1}$$

$$\delta = \frac{1}{2}\,\sqrt{2}e^{\left(-\lambda\left(x_3, x_4\right)\right)}\,\partial_{x_3} + \frac{1}{2}i\,\sqrt{2}e^{\left(-\nu\left(x_3, x_4\right)\right)}\,\partial_{x_4}$$

$$\overline{\delta} = \frac{1}{2}\,\sqrt{2}e^{\left(-\lambda\left(x_3, x_4\right)\right)}\,\partial_{x_3} - \frac{1}{2}i\,\sqrt{2}e^{\left(-\nu\left(x_3, x_4\right)\right)}\,\partial_{x_4}$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = 0$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = \frac{1}{4} i\,\sqrt{2} e^{\left(-\nu\left(x_3, x_4\right)\right)} \frac{\partial}{\partial x_4} \lambda\left(x_3, x_4\right) - \frac{1}{4}\,\sqrt{2} e^{\left(-\lambda\left(x_3, x_4\right)\right)} \frac{\partial}{\partial x_3} \nu\left(x_3, x_4\right)$$

$$\beta = \frac{1}{4} i\,\sqrt{2} e^{\left(-\nu\left(x_3, x_4\right)\right)} \frac{\partial}{\partial x_4} \lambda\left(x_3, x_4\right) + \frac{1}{4}\,\sqrt{2} e^{\left(-\lambda\left(x_3, x_4\right)\right)} \frac{\partial}{\partial x_3} \nu\left(x_3, x_4\right)$$

$$\tau = -\frac{\sqrt{2} e^{\left(-\lambda\left(x_3, x_4\right)\right)} \frac{\partial}{\partial x_3} Y\left(x_3, x_4\right)}{2\,Y\left(x_3, x_4\right)} - \frac{i\,\sqrt{2} e^{\left(-\nu\left(x_3, x_4\right)\right)} \frac{\partial}{\partial x_4} Y\left(x_3, x_4\right)}{2\,Y\left(x_3, x_4\right)}$$

$$\pi = \frac{\sqrt{2} e^{\left(-\lambda\left(x_3, x_4\right)\right)} \frac{\partial}{\partial x_3} Y\left(x_3, x_4\right)}{2\,Y\left(x_3, x_4\right)} - \frac{i\,\sqrt{2} e^{\left(-\nu\left(x_3, x_4\right)\right)} \frac{\partial}{\partial x_4} Y\left(x_3, x_4\right)}{2\,Y\left(x_3, x_4\right)}$$

$$\epsilon = -\frac{\sqrt{2}}{4\,x_1 Y\left(x_3, x_4\right)}$$

$$\gamma = -\frac{\sqrt{2}}{4\,x_1 Y\left(x_3, x_4\right)}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\begin{aligned}
\Phi_{02} = {} & \frac{e^{(-2\,\lambda(x_3,x_4))}\,\frac{\partial}{\partial x_3}Y\left(x_3,x_4\right)\,\frac{\partial}{\partial x_3}\lambda\left(x_3,x_4\right)}{2\,Y\left(x_3,x_4\right)} + \frac{i\,e^{(-\lambda(x_3,x_4)-\nu(x_3,x_4))}\,\frac{\partial}{\partial x_3}Y\left(x_3,x_4\right)\,\frac{\partial}{\partial x_4}\lambda\left(x_3,x_4\right)}{Y\left(x_3,x_4\right)} - \frac{e^{(-2\,\nu(x_3,x_4))}\,\frac{\partial}{\partial x_4}Y\left(x_3,x_4\right)\,\frac{\partial}{\partial x_4}\lambda\left(x_3,x_4\right)}{2\,Y\left(x_3,x_4\right)} \\
& + \frac{e^{(-2\,\lambda(x_3,x_4))}\,\frac{\partial}{\partial x_3}Y\left(x_3,x_4\right)\,\frac{\partial}{\partial x_3}\nu\left(x_3,x_4\right)}{2\,Y\left(x_3,x_4\right)} + \frac{i\,e^{(-\lambda(x_3,x_4)-\nu(x_3,x_4))}\,\frac{\partial}{\partial x_4}Y\left(x_3,x_4\right)\,\frac{\partial}{\partial x_3}\nu\left(x_3,x_4\right)}{Y\left(x_3,x_4\right)} - \frac{e^{(-2\,\nu(x_3,x_4))}\,\frac{\partial}{\partial x_4}Y\left(x_3,x_4\right)\,\frac{\partial}{\partial x_4}\nu\left(x_3,x_4\right)}{2\,Y\left(x_3,x_4\right)} \\
& - \frac{e^{(-2\,\lambda(x_3,x_4))}\,\frac{\partial^2}{(\partial x_3)^2}Y\left(x_3,x_4\right)}{2\,Y\left(x_3,x_4\right)} - \frac{i\,e^{(-\lambda(x_3,x_4)-\nu(x_3,x_4))}\,\frac{\partial^2}{\partial x_3\partial x_4}Y\left(x_3,x_4\right)}{Y\left(x_3,x_4\right)} + \frac{e^{(-2\,\nu(x_3,x_4))}\,\frac{\partial^2}{(\partial x_4)^2}Y\left(x_3,x_4\right)}{2\,Y\left(x_3,x_4\right)}
\end{aligned}$$

$$\Phi_{10} = 0$$

$$\begin{aligned}
\Phi_{11} = {} & -\frac{1}{4}\,e^{(-2\,\nu(x_3,x_4))}\,\frac{\partial}{\partial x_4}\lambda\left(x_3,x_4\right)^2 + \frac{1}{4}\,e^{(-2\,\lambda(x_3,x_4))}\,\frac{\partial}{\partial x_3}\lambda\left(x_3,x_4\right)\,\frac{\partial}{\partial x_3}\nu\left(x_3,x_4\right) - \frac{1}{4}\,e^{(-2\,\lambda(x_3,x_4))}\,\frac{\partial}{\partial x_3}\nu\left(x_3,x_4\right)^2 + \frac{1}{4}\,e^{(-2\,\nu(x_3,x_4))}\,\frac{\partial}{\partial x_4}\lambda\left(x_3,x_4\right)\,\frac{\partial}{\partial x_4}\nu\left(x_3,x_4\right) \\
& - \frac{1}{4}\,e^{(-2\,\nu(x_3,x_4))}\,\frac{\partial^2}{(\partial x_4)^2}\lambda\left(x_3,x_4\right) - \frac{1}{4}\,e^{(-2\,\lambda(x_3,x_4))}\,\frac{\partial^2}{(\partial x_3)^2}\nu\left(x_3,x_4\right) + \frac{e^{(-2\,\lambda(x_3,x_4))}\,\frac{\partial}{\partial x_3}Y\left(x_3,x_4\right)^2}{4\,Y\left(x_3,x_4\right)^2} + \frac{e^{(-2\,\nu(x_3,x_4))}\,\frac{\partial}{\partial x_4}Y\left(x_3,x_4\right)^2}{4\,Y\left(x_3,x_4\right)^2}
\end{aligned}$$

$$\Phi_{12} = 0$$

$$\begin{aligned}
\Phi_{20} = {} & \frac{e^{(-2\,\lambda(x_3,x_4))}\,\frac{\partial}{\partial x_3}Y\left(x_3,x_4\right)\,\frac{\partial}{\partial x_3}\lambda\left(x_3,x_4\right)}{2\,Y\left(x_3,x_4\right)} - \frac{i\,e^{(-\lambda(x_3,x_4)-\nu(x_3,x_4))}\,\frac{\partial}{\partial x_3}Y\left(x_3,x_4\right)\,\frac{\partial}{\partial x_4}\lambda\left(x_3,x_4\right)}{Y\left(x_3,x_4\right)} - \frac{e^{(-2\,\nu(x_3,x_4))}\,\frac{\partial}{\partial x_4}Y\left(x_3,x_4\right)\,\frac{\partial}{\partial x_4}\lambda\left(x_3,x_4\right)}{2\,Y\left(x_3,x_4\right)} \\
& + \frac{e^{(-2\,\lambda(x_3,x_4))}\,\frac{\partial}{\partial x_3}Y\left(x_3,x_4\right)\,\frac{\partial}{\partial x_3}\nu\left(x_3,x_4\right)}{2\,Y\left(x_3,x_4\right)} - \frac{i\,e^{(-\lambda(x_3,x_4)-\nu(x_3,x_4))}\,\frac{\partial}{\partial x_4}Y\left(x_3,x_4\right)\,\frac{\partial}{\partial x_3}\nu\left(x_3,x_4\right)}{Y\left(x_3,x_4\right)} - \frac{e^{(-2\,\nu(x_3,x_4))}\,\frac{\partial}{\partial x_4}Y\left(x_3,x_4\right)\,\frac{\partial}{\partial x_4}\nu\left(x_3,x_4\right)}{2\,Y\left(x_3,x_4\right)} \\
& - \frac{e^{(-2\,\lambda(x_3,x_4))}\,\frac{\partial^2}{(\partial x_3)^2}Y\left(x_3,x_4\right)}{2\,Y\left(x_3,x_4\right)} + \frac{i\,e^{(-\lambda(x_3,x_4)-\nu(x_3,x_4))}\,\frac{\partial^2}{\partial x_3\partial x_4}Y\left(x_3,x_4\right)}{Y\left(x_3,x_4\right)} + \frac{e^{(-2\,\nu(x_3,x_4))}\,\frac{\partial^2}{(\partial x_4)^2}Y\left(x_3,x_4\right)}{2\,Y\left(x_3,x_4\right)}
\end{aligned}$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = 0$$

$$
\begin{aligned}
\Lambda = & -\frac{1}{12}\,e^{\left(-2\,\nu(x_3,x_4)\right)}\frac{\partial}{\partial x_4}\lambda\left(x_3,x_4\right)^2+\frac{1}{12}\,e^{\left(-2\,\lambda(x_3,x_4)\right)}\frac{\partial}{\partial x_3}\lambda\left(x_3,x_4\right)\frac{\partial}{\partial x_3}\nu\left(x_3,x_4\right)-\frac{1}{12}\,e^{\left(-2\,\lambda(x_3,x_4)\right)}\frac{\partial}{\partial x_3}\nu\left(x_3,x_4\right)^2+\frac{1}{12}\,e^{\left(-2\,\nu(x_3,x_4)\right)}\frac{\partial}{\partial x_4}\lambda\left(x_3,x_4\right)\frac{\partial}{\partial x_4}\nu\left(x_3,x_4\right)\\
& +\frac{e^{\left(-2\,\lambda(x_3,x_4)\right)}\frac{\partial}{\partial x_3}Y\left(x_3,x_4\right)\frac{\partial}{\partial x_3}\lambda\left(x_3,x_4\right)}{6\,Y\left(x_3,x_4\right)}-\frac{e^{\left(-2\,\nu(x_3,x_4)\right)}\frac{\partial}{\partial x_4}Y\left(x_3,x_4\right)\frac{\partial}{\partial x_4}\lambda\left(x_3,x_4\right)}{6\,Y\left(x_3,x_4\right)}-\frac{1}{12}\,e^{\left(-2\,\nu(x_3,x_4)\right)}\frac{\partial^2}{(\partial x_4)^2}\lambda\left(x_3,x_4\right)\\
& -\frac{e^{\left(-2\,\lambda(x_3,x_4)\right)}\frac{\partial}{\partial x_3}Y\left(x_3,x_4\right)\frac{\partial}{\partial x_3}\nu\left(x_3,x_4\right)}{6\,Y\left(x_3,x_4\right)}-\frac{1}{12}\,e^{\left(-2\,\lambda(x_3,x_4)\right)}\frac{\partial^2}{(\partial x_3)^2}\nu\left(x_3,x_4\right)+\frac{e^{\left(-2\,\nu(x_3,x_4)\right)}\frac{\partial}{\partial x_4}Y\left(x_3,x_4\right)\frac{\partial}{\partial x_4}\nu\left(x_3,x_4\right)}{6\,Y\left(x_3,x_4\right)}\\
& -\frac{e^{\left(-2\,\lambda(x_3,x_4)\right)}\frac{\partial}{\partial x_3}Y\left(x_3,x_4\right)^2}{12\,Y\left(x_3,x_4\right)^2}-\frac{e^{\left(-2\,\lambda(x_3,x_4)\right)}\frac{\partial^2}{(\partial x_3)^2}Y\left(x_3,x_4\right)}{6\,Y\left(x_3,x_4\right)}-\frac{e^{\left(-2\,\nu(x_3,x_4)\right)}\frac{\partial}{\partial x_4}Y\left(x_3,x_4\right)^2}{12\,Y\left(x_3,x_4\right)^2}-\frac{e^{\left(-2\,\nu(x_3,x_4)\right)}\frac{\partial^2}{(\partial x_4)^2}Y\left(x_3,x_4\right)}{6\,Y\left(x_3,x_4\right)}
\end{aligned}
$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$
\begin{aligned}
\Psi_2 = & \frac{1}{6}\,e^{\left(-2\,\nu(x_3,x_4)\right)}\frac{\partial}{\partial x_4}\lambda\left(x_3,x_4\right)^2-\frac{1}{6}\,e^{\left(-2\,\lambda(x_3,x_4)\right)}\frac{\partial}{\partial x_3}\lambda\left(x_3,x_4\right)\frac{\partial}{\partial x_3}\nu\left(x_3,x_4\right)+\frac{1}{6}\,e^{\left(-2\,\lambda(x_3,x_4)\right)}\frac{\partial}{\partial x_3}\nu\left(x_3,x_4\right)^2-\frac{1}{6}\,e^{\left(-2\,\nu(x_3,x_4)\right)}\frac{\partial}{\partial x_4}\lambda\left(x_3,x_4\right)\frac{\partial}{\partial x_4}\nu\left(x_3,x_4\right)\\
& +\frac{e^{\left(-2\,\lambda(x_3,x_4)\right)}\frac{\partial}{\partial x_3}Y\left(x_3,x_4\right)\frac{\partial}{\partial x_3}\lambda\left(x_3,x_4\right)}{6\,Y\left(x_3,x_4\right)}-\frac{e^{\left(-2\,\nu(x_3,x_4)\right)}\frac{\partial}{\partial x_4}Y\left(x_3,x_4\right)\frac{\partial}{\partial x_4}\lambda\left(x_3,x_4\right)}{6\,Y\left(x_3,x_4\right)}+\frac{1}{6}\,e^{\left(-2\,\nu(x_3,x_4)\right)}\frac{\partial^2}{(\partial x_4)^2}\lambda\left(x_3,x_4\right)\\
& -\frac{e^{\left(-2\,\lambda(x_3,x_4)\right)}\frac{\partial}{\partial x_3}Y\left(x_3,x_4\right)\frac{\partial}{\partial x_3}\nu\left(x_3,x_4\right)}{6\,Y\left(x_3,x_4\right)}+\frac{1}{6}\,e^{\left(-2\,\lambda(x_3,x_4)\right)}\frac{\partial^2}{(\partial x_3)^2}\nu\left(x_3,x_4\right)+\frac{e^{\left(-2\,\nu(x_3,x_4)\right)}\frac{\partial}{\partial x_4}Y\left(x_3,x_4\right)\frac{\partial}{\partial x_4}\nu\left(x_3,x_4\right)}{6\,Y\left(x_3,x_4\right)}\\
& +\frac{e^{\left(-2\,\lambda(x_3,x_4)\right)}\frac{\partial}{\partial x_3}Y\left(x_3,x_4\right)^2}{6\,Y\left(x_3,x_4\right)^2}-\frac{e^{\left(-2\,\lambda(x_3,x_4)\right)}\frac{\partial^2}{(\partial x_3)^2}Y\left(x_3,x_4\right)}{6\,Y\left(x_3,x_4\right)}+\frac{e^{\left(-2\,\nu(x_3,x_4)\right)}\frac{\partial}{\partial x_4}Y\left(x_3,x_4\right)^2}{6\,Y\left(x_3,x_4\right)^2}-\frac{e^{\left(-2\,\nu(x_3,x_4)\right)}\frac{\partial^2}{(\partial x_4)^2}Y\left(x_3,x_4\right)}{6\,Y\left(x_3,x_4\right)}
\end{aligned}
$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"